Sharp linear bounds for the EKG sequence

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Abstract

The EKG sequence starts $a(1) = 1$, $a(2) = 2$, and thereafter takes the least unused positive integer sharing a factor greater than 1 with its predecessor. In 2002 Lagarias, Rains and Sloane conjectured that $13n/28 \le a(n) \le 12n/7$ for every $n \ge 1$, with equality only at $n = 28$ and $n = 7$ respectively. We prove this conjecture. Earlier work gives coarse linear bounds at every n (Lagarias, Rains and Sloane), the limits 1/2, 3/2 and 1 of $a(n)/n$ on primes, their triples and the other terms (Hofman and Pilipczuk), and explicit rates (Haver), but not the sharp constants at every n. Our main step bounds the number of used even values greater than the least unused even number E by a self-improving inequality closed with the large sieve. Explicit estimates then hold once E exceeds about $1.3 \cdot 10^{10}$, which happens within the first $1.6 \cdot 10^{10}$ terms; those terms are computed. Lean 4 checks the deduction from the definition, conditional on computed premises; separate programs check those premises, and their match with the Lean statements was checked by reading the code.

AI production. The proofs, the Lean formalization, the programs and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 Introduction

The EKG sequence, introduced by Lagarias, Rains and Sloane [7], is A064413 in the OEIS [13]. Write $U_t = \{a(1), \dots, a(t)\}$. The sequence is defined by

$$a(1) = 1, \quad a(2) = 2, \quad a(n) = \min\{m \ge 1 : \ m \notin U_{n-1}, \ \gcd(m, a(n-1)) > 1\} \quad (n \ge 3), \tag{1}$$

so it begins $1, 2, 4, 6, 3, 9, 12, 8, 10, 5, 15, 18, 14, 7, 21, 24, 16, 20, 22, 11, \dots$. It is a permutation of the positive integers [7, Theorem 1] (Theorem 2.5 in the journal version). Its graph has three visible lines (Figure 1): most terms satisfy $a(n) \approx n$, while each odd prime p is immediately preceded by $2p$ and followed by $3p$, and when $a(n) = p$ we have $p \approx n/2$ (and $a(n+1) = 3p \approx 3n/2$). Lagarias, Rains and Sloane proved $n/260 \le a(n) \le 14n$ and conjectured sharp linear bounds [7, Conjecture 3, end of §4] (Conjecture 4.3 in the journal version):

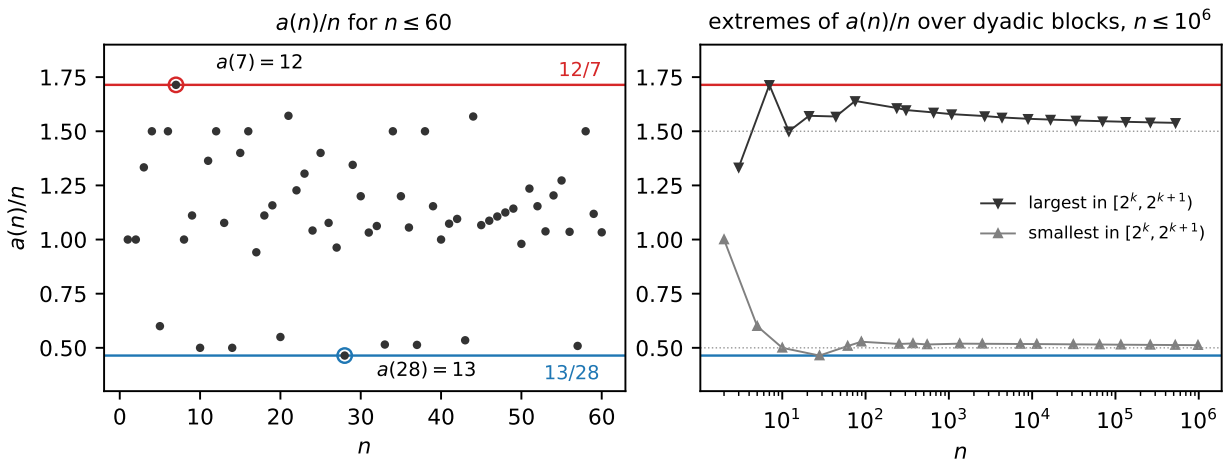


Figure 1: Left: $a(n)/n$ for $n \le 60$, with the bounds $12/7$ (top) and $13/28$ (bottom) of Theorem 1, attained at $n = 7$ and $n = 28$. Right: the largest and smallest values of $a(n)/n$ over each block $2^k \le n < 2^{k+1}$, $n \le 10^6$, approaching $3/2$ and $1/2$ (dotted). Drawn from the computed sequence by `data/figure.py`.

> *The sequence $a(n)$ satisfies $a(n) \ge \frac{13}{28}n$, with equality if and only if $n = 28$, and $a(n) \le \frac{12}{7}n$, with equality if and only if $n = 7$.*

Theorem 1. *For every integer $n \ge 1$,*

$$\frac{13}{28}\,n \;\le\; a(n) \;\le\; \frac{12}{7}\,n.$$

The lower bound is an equality only for $n = 28$ (where $a(28) = 13$) and the upper bound only for $n = 7$ (where $a(7) = 12$).

Prior work. Lagarias, Rains and Sloane [7] defined the sequence, proved the linear bounds above and the rule that $a(n)$ is the least unused multiple of some prime factor of $a(n-1)$, and conjectured both the sharp bounds of Theorem 1 and the asymptotics $a(n) \sim n$, $a(n) \sim n/2$ and $a(n) \sim 3n/2$ on the three lines. Hofman and Pilipczuk [5] proved that every odd prime p is immediately preceded by $2p$, bounded the crossings of a level by composite terms, and proved the asymptotics. Haver [4] made these arguments explicit and formalized them in Lean: he completes the pattern $2p, p, 3p$, reduces its computed base to the primes $p \le 43$, bounds the number of used even numbers above the least missing even number, the spread of the terms above it, and the unused composites below it, and relates the index to that even number by a counting identity. The author [3] then obtained a second-order asymptotic, $n + n/(3\log n) + o(n/\log n)$, for a smoothed version of the sequence (each p and $3p$ replaced by $2p$), on Haver's Lean foundation. The present proof uses Haver's Lean definition of the sequence and his estimates, and the author's exact form of the counting identity, as proved Lean theorems.

What is new, and the hardest step. The bounds of Lagarias, Rains and Sloane hold at every n but are far from sharp; the later results are asymptotic, and the explicit forms apply only from enormous thresholds. Theorem 1 needs the sharp constants at every n, and its margin is small: the lower line $a(n)/n \to 1/2$ sits only $1/28$ above $13/28$. The obstacle is the number K of *early evens*: at a given moment, the even numbers already used that exceed the least even number E not yet used. With Haver's bound $K \le 12\pi(\sqrt{E}) + 720$ [4, Theorem 5] the argument closes only for very large n; we use it that way for $n \ge 10^{156}$ (Lemma 8), far beyond any computation. The new step is a self-improving bound (Lemma 2, Corollary 3): the primes responsible for the excess of K lie among the $3K+2$ integers just below $E/2$, so $K \le \pi(\sqrt{E}) + 2 + \#\{\text{primes among } 3K+2 \text{ consecutive integers}\}$, and a Brun–Titchmarsh bound from the large sieve turns this into $K \le (1+o(1))\pi(\sqrt{E})$. The bound is proved by induction over the moments of the sequence, so it needs no prior bound on K: at each moment the earlier ones already satisfy it, and an explicit inequality, checked on intervals of E, carries it to the next. At $E = 1.3328 \cdot 10^{10}$ the certified bound is 28,017, against 131,628 from Haver's bound. The hardest part is making every constant small enough for these inequalities to close at a size of E a computation can reach. A second large-sieve bound controls the odd terms above E (Lemma 5). With these, explicit estimates close once $E \ge 1.3328 \cdot 10^{10}$; the computed first $1.6 \cdot 10^{10}$ terms reach that point (with E above $1.41 \cdot 10^{10}$ by $n = 1.393 \cdot 10^{10}$) and settle every smaller n.

How it is checked. Theorem 1 is a Lean theorem [2, 9] about Haver's definition of the sequence, conditional on four computed premises: three facts about the first $1.6 \cdot 10^{10}$ terms and the existence of a cover of $[1.3328 \cdot 10^{10}, 10^{205})$ by intervals satisfying explicit inequalities. Every analytic input (explicit prime counting bounds and the large sieve) is proved in Lean. The premises are computed, and each is checked by programs that share no code with the ones that produced it; that the programs compute what the premises state was checked by reading the code. Section 9 lists exactly what is trusted.

Outline. Section 2 fixes notation and recalls the known facts. Section 3 gives the strategy. Sections 4–6 prove the middle range, Section 7 the far range and Section 8 describes the computed range. Section 9 describes the formalization and the checks.

2 Notation and known facts

A *state* $t \ge 1$ is the moment after $a(1), \dots, a(t)$ have been placed. For a state t define:

- $E(t)$, the *frontier*: the least positive even integer not in U_t. It is nondecreasing in t, and every even number below $E(t)$ is used.
- $K(t)$, the number of *early evens*: even $w \in U_t$ with $w > E(t)$.
- $B_q(t)$, the least multiple of q not in U_t.

- A *tick value* is $3p$ with p an odd prime. $T(t)$ is the largest element of U_t that is not a tick value, and $O(t)$ is the number of odd $w \in U_t$ with $w > E(t)$ that are not tick values.
- $H(t)$, the number of *holes*: composite $h < E(t)$ with $h \notin U_t$.
- $A(x) = \pi(x) - 2\pi(x/2) + \pi(x/3)$, where π counts primes.

For example, after $a(1..7) = 1, 2, 4, 6, 3, 9, 12$ we have $E(7) = 8$, the early even 12, so $K(7) = 1$, and $O(7) = 0$ since 9 is a tick value.

We use the following facts; each is a Lean theorem in the foundation (Section 9).

(F1) *Pointer rule* [7]. $a(n) = \min\{B_q(n-1) : q \text{ prime},\ q \mid a(n-1)\}$ for $n \geq 3$.

(F2) *Prime pattern.* For each odd prime p there is an index N_p with $a(N_p) = 2p = E(N_p - 1)$, $a(N_p + 1) = p$ and $a(N_p + 2) = 3p$; the value $3p$ occurs only at $N_p + 2$, and indices N_p of distinct primes differ by at least 4. That p follows the first multiple of p is [7, Lemma 1] (Lemma 2.1 in the journal version), and that this multiple is $2p$ is [5, Theorem 4.3]. Hofman and Pilipczuk prove the latter by a crossing count at the level $2p$ that closes for $p > 25000$, and cover smaller p by the computation of 10^7 terms in [7]; Haver [4, Remark 3] closes the same count for every $p > 43$, leaving $p \leq 43$, which the first 82 terms show. The rest, $2p = E(N_p - 1)$, the term $3p$, its uniqueness and the spacing, is proved in [4, (F2)]. In the Lean proof the primes $p \leq 43$ are checked on the first terms, $43 < p < 4096$ by a finite computation, and larger p by explicit prime bounds.

(F3) *Crossings* [5, Lemma 4.5], [4]. For a level $x > 2$, at most $\pi(\sqrt{x})$ indices n have $a(n)$ composite and $a(n) < x \leq a(n+1)$.

(L7) [4, Lemma 7] $B_3(t) \leq E(t) + 6K(t) + 6$ for $t \geq 9$.

(SP) *Span* [4, Theorem 8]. With $k = K(t) + 1$ and $e = E(t)$, $T(t) - e \leq 2k^2 + 2k\sqrt{k^2 + e}$ for $t \geq 9$.

(EE) Our variant of [4, Lemma 4], from its proof. Call $i < t$ a *prime up* at state t if $a(i)$ is prime and $a(i) < E(t) \leq a(i+1)$; it is *paired* if $a(i+1)$ is odd and $a(i+2) < E(t)$, and *unpaired* otherwise. Then $K(t) \leq \pi(\sqrt{E(t)}) + 1 + \#\{\text{unpaired prime ups at } t\}$.

(CL) *Counting identity* [4], in the exact form of [3]. For $t \geq 9$ and $e = E(t)$,

$$e - t = A(e) + H(t) - K(t) - O(t) + c(t), \qquad 1 \leq c(t) \leq 5.$$

The analytic inputs are explicit and all are proved in Lean. Put $c = 1.03883$, $\alpha = 0.92129$ and, for $x \geq 256$, $F(x) = \frac{x}{\log x}\left(1 + \frac{1}{\log x} + \frac{2}{\log^2 x} + \frac{24}{\log^3 x}\right)$, $R(x) = c\sqrt{x} + \frac{c}{\log 2} x^{1/3} \log x$.

(P) *Prime counting.* For $x \geq 256$, $\pi^-(x) \leq \pi(x) \leq \pi^+(x)$, where $\pi^+(x) = cF(x) + J$ with $J = \pi(256) - \theta(256)/\log 256 + c\,(256/\log 256 - F(256))$, where $\pi(256) = 54$ and $\theta(256) = \sum_{p<256} \log p$, and $\pi^-(x) = \max\bigl(0, (\alpha x - 5\log x + 5 - R(x))/\log x\bigr)$; below 256 both equal $\pi(x)$. They follow by partial summation from Rosser and Schoenfeld's $\psi(x) < cx$ [14, Theorem 12] and $\psi(x) \geq \alpha x - 5\log x + 5$ for $x \geq 30$, a weaker form of Chebyshev's $\psi(x) > Ax - \frac{5}{2}\log x - 1$, $A = 0.92129\ldots$ [1, §V], as proved in Lean in [6].

(BT) *Primes in short intervals.* For real $x > 0$, $y \geq 1$ and integers $Q \geq 6$,

$$\pi(x+y) - \pi(x) \leq \frac{y + Q^2}{\log(Q+1) + 1.07} + \pi(Q).$$

This is the large sieve [10, 11] with $G(Q) = \sum_{d \leq Q} \mu^2(d)/\varphi(d) > \log(Q+1) + 1.07$, the form in the proof of [11, Lemma 7], which states $G(z) \geq \log z + 1.07$ for real $z \geq 6$.

(LS) *Arithmetic large sieve* [10, 11]. A set of integers in an interval of N consecutive integers that avoids one residue class modulo every prime $p \leq Q$ has at most $(N + Q^2)/\log Q$ elements.

3 Strategy

The three kinds of terms behave differently, and each is located through the frontier. If $a(n) = p$ is an odd prime then $n = N_p + 1$ and $E(n-2) = 2p$ by (F2); if $a(n) = 3p$ then $n = N_p + 2$ and $E(n-3) = 2p$; every other term either lies above the frontier $E(n-1)$, and then at most $T(n)$, or fills a hole below it, and holes lie close to the frontier. The counting identity (CL) relates the index t to $e = E(t)$: $A(e) \approx e/(3\log e)$ is small, and if K, O and H are small compared to e, then $t \approx e$ and the three kinds of terms give $a(n)/n$ near $1/2$, $3/2$ and 1.

Everything therefore rests on explicit upper bounds for K, O and H at every state, uniformly in a range of e. Haver bounds each by a multiple of $\pi(\sqrt{e})$ (times further factors), so the bounds are relatively small only when e is astronomically large. We split the indices into three ranges:

1. $n \leq L = 1.6 \cdot 10^{10}$: computed (Section 8);
2. $L < n < 10^{203}$: a bootstrap bound $K(t) \leq M(E(t))$ for a step function M, certified with $E(L-2) \geq e_0 = 1.3328 \cdot 10^{10}$ and on every interval of a cover of $[e_0, 10^{205})$ (Sections 4–6);
3. $n \geq 10^{156}$: Haver's bounds alone (Section 7).

The ranges overlap, so every n is covered. The hardest step is the bound on K in range 2, which must be good enough already at $e = 1.3 \cdot 10^{10}$.

4 Early evens

This section proves the self-improving bound on K. For a state t let $K^<(t) = \max\{K(s) : 1 \le s < t\}$.

Lemma 2 (Lean: `EKG3.lemmaA`). *Let t be a state and $e = E(t)$. Every unpaired prime up i at state t has $p = a(i)$ prime with $2p < e < 2p + 6K^<(t) + 6$.*

Proof. The prime 2 is never a prime up: $a(2) = 2$, $a(3) = 4$, and $E(t) \ge 6$ for $t \ge 3$. So p is odd, and by (F2) $a(i-1) = 2p = E(i-2)$ and $a(i+1) = 3p$. Since $2p \in U_t$, $2p \ne e$, and E is nondecreasing, so $2p < e$. Unpairedness forces every multiple of 3 below e to be used by the state $k = i+1$ when $k < t$ (otherwise $a(i+2) \le B_3(k) < e$ by (F1), since $3 \mid a(k)$); in particular $3 \nmid e$, as e is still unused at t. Each multiple w of 6 with $2p < w < e$ is then used by state k, is none of $2p$, p, $3p$, and so was already used at state $i-2$, where $E(i-2) = 2p < w$: it is an early even at state $i-2 < t$. Since $6 \nmid e$ there are more than $(e-2p)/6 - 1$ such w, so $(e-2p)/6 - 1 < K(i-2) \le K^<(t)$. In the remaining case $i+1 = t$ we have $U_t = U_{i-2} \cup \{2p, p, 3p\}$, so every even w with $2p < w < e$ lies in U_{i-2} and is an early even at state $i-2$; hence $(e-2p)/2 - 1 \le K^<(t)$, which is stronger. □

Corollary 3 (Bootstrap; Lean: `EKG3.boundedCorollaryA`). *Let $e_0 < E_1$, let M be a nondecreasing function on the integers, and let $\mathrm{BT}(y)$ bound the number of primes in every block of y consecutive integers. Suppose that $K(s) \le M(E(s))$ at every state s with $E(s) < e_0$, and that*

$$\pi(\sqrt{e}) + 2 + \mathrm{BT}(3M(e) + 2) \le M(e) \tag{2}$$

for every even e with $e_0 \le e < E_1$. Then $K(t) \le M(E(t))$ at every state t with $E(t) < E_1$.

Proof. Strong induction on t. The states $t \le 9$ have $K(t) \le 1$ by direct computation of the first terms, and (2) gives $M(e) \ge 2$, so they satisfy the claim. Let $e = E(t) \ge e_0$. Every $s < t$ has $E(s) \le e$, so by induction and monotonicity $K^<(t) \le M(e)$. By Lemma 2 the primes p of the unpaired prime ups at t satisfy $e/2 - 3M(e) - 3 < p < e/2$, so they lie among $3M(e) + 2$ consecutive integers, and they are distinct because the sequence does not repeat values. Thus (EE) gives $K(t) \le \pi(\sqrt{e}) + 1 + \mathrm{BT}(3M(e) + 2) \le M(e)$ by (2). □

With (BT), $\mathrm{BT}(y) \le (y + Q^2)/(\log(Q+1) + 1.07) + \pi(Q)$ for any integer $Q \ge 6$. For $y = 3M + 2$ and Q near $\sqrt{y/\log y}$ this is about $2y/\log y \approx 6M/\log M$, so (2) holds once M is a little larger than $\pi(\sqrt{e})/(1 - 6/\log M)$. This is $(1 + o(1))\pi(\sqrt{e})$ as $e \to \infty$, and about $2.6\,\pi(\sqrt{e})$ at $e = 1.3328 \cdot 10^{10}$. Haver's argument counted the same primes through their residues modulo 210, which bounds them only by a fixed fraction of the window; this is where his factor 12 comes from.

5 Holes and odd terms above the frontier

The next two lemmas bound H and O in terms of a bound M on K. Lemma 4 is Haver's hole bound [4, Theorem 11] with an arbitrary nondecreasing bound M in place of his explicit one, and with the hypothesis needed only at earlier states.

Lemma 4 (Holes; Lean: `EKG3.lemmaC`). *Let M be nondecreasing with $K(s) \le M(E(s))$ for every state $s < t$, and $e = E(t)$. Every hole $h \ge 15$ at state t satisfies $e - h \le 2\,\mathrm{spf}(h)\Big(\pi(\sqrt{h}) + M(h) + 4\Big)$, where $\mathrm{spf}(h)$ is the least prime factor of h, and $H(t) \le \pi(\sqrt{e})\Big(\pi(\sqrt{e}) + M(e) + 5\Big)$.*

Sketch. A hole h is odd; put $s = \mathrm{spf}(h)$ and let t_1 be the first state with $E(t_1) > h$. Every multiple w of $2s$ in (h, e) is used. If it was placed after t_1 then the next term is at most $B_s < h$ by (F1), a down-crossing of the level h; at most $K(t_1 - 1) \le M(h)$ of them were placed before t_1. Up-crossings come from composite bases (at most $\pi(\sqrt{h})$ by (F3)) or from primes p with $a(n) = 3p$, each of which is matched with its own down-crossing $2p \to p$. As ups and downs alternate, the number of multiples of $2s$ in (h, e) is at most $\pi(\sqrt{h}) + M(h) + 3$, which gives the first bound. Summing over the primes $s \le \sqrt{e}$ the number of odd multiples of s in the resulting window gives the second. □

Lemma 5 (Odd terms above the frontier; Lean: `EKG3.lemmaD`). *Let $t \ge 10$ be a state, $e = E(t)$, $K' = \max_{s \le t} K(s)$, and let $Y \ge 6K' + 7$ be an integer. Put $X = e + Y + \frac{1}{2}$ and $R = (Y + \frac{1}{2})/(2(K(t) + 1))$. Then*

$$O(t) \le \left\lceil \frac{Y}{2} \right\rceil + \pi(\sqrt{X}) + Z,$$

where Z is the number of integers m with $X < m \le T(t)$ all of whose prime factors exceed R.

Proof. At most $\lceil Y/2 \rceil$ odd values lie in (e, X). Let $v = a(i) > X$ be another value counted by $O(t)$, and $c = a(i - 1)$ its predecessor. If c were even, (F1) with $q = 2$ would give $v \le E(i - 1) \le e$; if c were 1 or an odd prime, v would be 2 or a tick value (F2); if c were a tick value, (F1) with $q = 3$ and (L7) would give $v \le e + 6K' + 6 < X$. So c is an odd composite, not a tick value, with $c \le T(t)$. For each prime $r \mid c$, $B_r(i - 1) \ge v$ by (F1), so the multiples of $2r$ in (e, v) are all early evens at t: there are at least $(v - e)/(2r) - 1$ of them, whence $r > R$. If $c < X$ the step $c \to v$ crosses the level X upward, which happens at most $\pi(\sqrt{X})$ times by (F3); otherwise c is counted by Z, and $v \mapsto c$ is injective. □

By (SP) the numbers counted by Z lie in an interval of at most $S - Y$ integers, $S = 2k^2 + 2k\sqrt{k^2 + e}$, and they avoid the residue 0 modulo every prime $p \le R$; so (LS) gives $Z \le (S - Y + Q^2)/\log Q$ for every integer $3 \le Q \le R$.

6 The middle range

We now combine the lemmas on a cover. Put $e_0 = 13{,}328{,}000{,}000$, $K_{\mathrm{fin}} = 329$, $E_1 = 10^{205}$ and $L = 16{,}000{,}000{,}000$.

Definition 6. A *row* is a tuple of nonnegative integers $(e_a, e_b, M, Y, Q, h_c, b_q)$. With $k = M+1$, $S = 2k^2 + 2k\sqrt{k^2+e_b}$, $p_U = \pi^+(\sqrt{e_b})$,

$$\begin{aligned} O^* &= \lfloor (Y+1)/2 \rfloor + \pi^+\Big(\sqrt{e_b + Y + 1}\Big) + (S - Y + Q^2)/\log Q, \\ H^* &= p_U(p_U + M + 5), \\ A_{\min} &= \pi^-(e_a) - 2\pi^+(e_b/2) + \pi^-(e_a/3), \\ A_{\max} &= \pi^+(e_b) - 2\pi^-(e_a/2) + \pi^+(e_b/3), \end{aligned}$$

the row is *valid* if $e_a < e_b$, $6M + 7 \le Y < S$, $3 \le Q \le (Y + \frac{1}{2})/(2(M+1))$, $h_c \ge 15$, $b_q \ge 6$ and

$$\begin{aligned} &\text{(B)} && p_U + 2 + \frac{3M + 2 + b_q^2}{\log(b_q+1) + 1.07} + \pi^+(b_q) \le M, \\ &\text{(Hc)} && h_c + 2\sqrt{h_c}\Big(\pi^+(\sqrt{h_c}) + M + 4\Big) < e_a, \\ &\text{(P)} && M + O^* + 1 - A_{\min} < e_a/13, \\ &\text{(Tk)} && A_{\max} + H^* + 2 < e_a/8, \\ &\text{(Od)} && 7S + 12(A_{\max} + H^* + 4) < 5e_a, \\ &\text{(Ol)} && 13(e_b - A_{\min} + M + O^*) < 28h_c. \end{aligned}$$

A *cover* is a list of valid rows with e_a of the first row equal to e_0, $M \ge K_{\text{fin}}$ in the first row, each e_a equal to the previous e_b, M nondecreasing, and last $e_b \ge E_1$.

In words: (B) is the bootstrap inequality (2) on the row; (Hc) puts every hole above h_c; (P) and (Tk) bound the index t above and below in terms of $E(t)$, which handles primes and tick values; (Od) bounds the terms above the frontier, and (Ol) the holes filled, for the remaining terms. O^*, H^*, $A_{\min}$ and $A_{\max}$ bound O, H and A on the row.

A cover exists: we found one with first row $M = 28{,}017$ and checked it as described in Section 9. Its tightest inequalities are (B), (Hc) and $Q \le (Y + \frac{1}{2})/(2(M+1))$, each satisfied with a ratio of about $1 - 10^{-9}$, and (Od), with ratio about $1 - 2 \cdot 10^{-6}$.

Theorem 7 (Lean: `EKG3.P3.theoremT`)**.** *Assume a cover exists, the bounds of Theorem 1 for $n \le L$ with its equality cases, $E(L-2) \ge e_0$, and $K(s) \le K_{\text{fin}}$ at every state s with $E(s) < e_0$. Then every n with $L < n < 10^{203}$ has $13n < 28a(n)$ and $7a(n) < 12n$. Moreover every state t with $e_0 \le E(t) < E_1$ has $t + 2 < 14E(t)/13$.*

Proof. Let $M(x) = K_{\text{fin}}$ for $x < e_0$ and $M(x) = M_j$ on the j-th row's interval $[e_a, e_b)$; it is nondecreasing. Condition (B) and (BT) with $Q = b_q$ give (2) on every row, so Corollary 3 yields $K(t) \le M(E(t))$ at every state with $E(t) < E_1$.

Fix a state t with $e = E(t)$ in the interval of row j. Then $K(s) \le M_j$ for all $s \le t$. Lemma 4 gives $H(t) \le H^*$, and every hole h exceeds h_c, because a hole $h \le h_c$ would give $e \le h_c + 2\sqrt{h_c}(\pi(\sqrt{h_c}) + M_j + 4) < e_a \le e$ by (Hc) (using $\text{spf}(h) \le \sqrt{h}$). (SP) gives

$T(t) - e \le S$, and Lemma 5 with Y and the sieve bound after it give $O(t) \le O^*$. Since $A_{\min} \le A(e) \le A_{\max}$, the identity (CL) yields

$$e - A_{\max} - H^* - 5 \;\le\; t \;\le\; e - A_{\min} + M + O^* - 1. \tag{3}$$

By (P), $t + 2 < e + e_a/13 \le 14e/13$; by (Tk), $t + 3 > 7e/8$.

Now let $L < n < 10^{203}$. Then $E(n) \le 2n + 2 < E_1$, and every state $t \ge n - 3$ has $E(t) \ge E(L - 2) \ge e_0$, so the bounds above apply at $t \in [n - 3, n]$.

- If $a(n) = p$ is an odd prime, take $t = n - 2$, so $e = 2p$. Then $13n = 13(t + 2) < 14e = 28p$, and $n > 7e/8 - 1$ gives $12n > 7p$.
- If $a(n) = 3p$, take $t = n - 3$, so $e = 2p$. Then $12n > 12 \cdot 7e/8 = 21p$, and $13n < 14e + 13 < 28 \cdot 3p$.
- Otherwise $v = a(n)$ is even or an odd composite that is not a tick value; take $t = n-1$. If $v > e$ then $v \le T(n) \le e+S$, and (3) with (Od) gives $7v < 12n$, while $13n < 14e < 28v$. If $v = e$ both bounds follow from $7e/8 < n + 2$ and $13n < 14e$. If $v < e$, then v is a hole at state t, so $v > h_c$ and (Ol) with (3) gives $28v > 13n$, while $7v < 7e < 12n$. □

7 The far range

For very large n Haver's explicit bounds suffice, without a cover.

Lemma 8 (Lean: `EKG3.P3.lemmaF`)**.** *Suppose every state t with $e_0 \le E(t) < E_1$ has $t+2 < 14E(t)/13$, and $E(L-2) \ge e_0$. Then every $n \ge 10^{156}$ has $13n < 28a(n)$ and $7a(n) < 12n$.*

Sketch. At the states $t \ge n - 3$ the hypothesis forces $E(t) > 9 \cdot 10^{155}$. From (P) one derives $0.921\,x/\log x \le \pi(x) \le 1.045\,x/\log x$ for $x \ge 10^{76}$. Inserted into Haver's bounds $K \le 12\pi(\sqrt{e}) + 720$ [4, Theorem 5], (SP), his hole bound and (CL) (with $O \le (T-e)/2+1$), this gives $0.997\,e < t$, $t+3 < 1.0766\,e$ and $T(t) < 1.153\,e$ at each such state t, with $e = E(t)$, and shows that every hole exceeds $e/1.1543$. Then the three cases of the proof of Theorem 7 give $a(n)/n$ between $1/(1.1543 \cdot 1.0766) > 0.8$ and $1.153/0.997 < 12/7$ for ordinary terms, and between $1/(2{\cdot}1.0766)$ and $3/(2{\cdot}0.997)$ for primes and tick values; the binding comparison is $13 \cdot 2 \cdot 1.0766 < 28$. □

8 The computed range

The premises of Theorem 7 about the first L terms are:

(Fa) for $1 \le n \le L$, $13n \le 28a(n)$ and $7a(n) \le 12n$, with equality only at $n = 28$ and $n = 7$;

(Fb) $E(L - 2) \ge e_0$;

(Fc) $K(s) \le 329$ at every state s with $E(s) < e_0$.

They are established by computing $a(1), \dots, a(L)$ with the pointer rule (F1), keeping one pointer per prime, and tracking E and K. We find $E(L) = 16{,}230{,}216{,}746$. The frontier first reaches $1.413 \cdot 10^{10}$ at $n = 13{,}928{,}388{,}828 < L-2$, which gives (Fb) since E is nondecreasing. The largest K at states with $E < 1.413 \cdot 10^{10}$ is 329; since every state with $E(s) < e_0$ comes before that crossing, this gives (Fc) for all states, not only those up to L. The only equality cases are $a(7) = 12$ and $a(28) = 13$.

Two programs written independently, with different factorization methods, memory layouts and formulas for K, compute these facts and agree; a third program checks the defining minimality property directly on the first 10^8 terms. Section 9 gives details.

9 Formalization and checks

The Lean theorem. The Lean statement `EKG3.Conj3Statement` says: for every $i \ge 0$, with $n = i + 1$ and $a(n) =$ `EKG.actualA` i, $13n \le 28a(n)$, $7a(n) \le 12n$, $13n = 28a(n) \iff n = 28$ and $7a(n) = 12n \iff n = 7$. These integer forms are equivalent to the rational ones. `EKG.actualA` is Haver's sequence: 1, 2, then the least positive unused integer sharing a prime factor with the previous term; it is proved to satisfy that specification (seeds, injectivity, the shared factor and minimality), and its first thirty terms are checked against the OEIS. The main theorem is

$$\texttt{EKG3.conj3_of_computations} :$$
$$\texttt{FiniteFa} \to \texttt{FiniteFbP3} \to \texttt{FiniteFcP3} \to \texttt{CertifiedCoverP3} \to \texttt{Conj3Statement},$$

whose premises are (Fa), (Fb), (Fc) and the existence of a cover (Definition 6), stated about the actual sequence with no free parameters. Lean checks the whole deduction of Sections 2–7, including (F1)–(CL) and the analytic inputs (P), (BT), (LS), and the build fails if any theorem uses an axiom beyond Lean's three standard ones.

The foundation is Haver's Lean archive [4] (`EKG-Lean4-sources-v2.zip`, deposited with his version 2 and retained unchanged in version 3) as extended in [3]. The analytic inputs reuse published formalizations: Rosser and Schoenfeld's Theorem 12 and a form of Chebyshev's lower bound from the IEANTN project [6] (with interval tools from LeanCert [8]), and the arithmetic large sieve from OpenAI's formalizations [12]. The bound $G(Q) > \log(Q+1) + 1.07$, the Brun–Titchmarsh form (BT) and the bounds (P) are proved here.

What is computed, and how it is checked. The four premises are not proved inside Lean; they are computed.

- (Fa), (Fb), (Fc) are produced by two programs that share no code: one keeps a smallest-prime-factor table and one pointer per prime, the other factors by segmented sieving with a paged cursor store and computes K by a different formula. Both run to $n = L$ and report the same equality cases, no violations, the same frontier thresholds, and

identical checkpoints every 10^8 terms: $(n, a(n), E, K, \max K)$ and a 64-bit hash of $a(1), \ldots, a(n)$. A third program, which uses neither pointers nor factorization, checks the defining property of every term up to 10^8 (each smaller multiple of a prime factor of $a(n-1)$ appears earlier); swapping two terms or changing one is detected both by it and by the hash.

- The cover is produced with interval arithmetic and checked by a separate program written from the Lean statement of Definition 6 alone, using ball arithmetic (Arb) at 256 bits and exact rationals and integers; every inequality is proved with the correct strictness. It accepts the cover and rejects corrupted ones (a gap, a decreased M, a Q above its bound, $b_q = 5$, $h_c = 14$, a moved start and a short end).

Trusted base. Beyond Lean's kernel, the proof trusts: the two prefix programs, their compilers and their recorded runs; the cover checker and the Arb library; and the correspondence between the programs and the Lean definitions, which was checked by reading rather than by machine: that both programs compute `EKG.actualA` by the pointer rule, that their E and K are Lean's `frontier` and `earlyEvenCount` at the matching state (the program's state n is Lean's state $n-1$), and that the program outputs give the premises as stated. The two prefix programs could in principle share a misreading of the definition; the direct minimality check covers only the first 10^8 terms.

Files. The Lean project, the programs, the certificate and the expected outputs accompany this paper. The Lean build takes about an hour on two cores, with up to about 7.5 GB of memory per Lean process. A quick check of the premises (minutes) checks the cover and its corruptions and replays the first 10^8 terms; the full replay of the first L terms takes about three hours and about 72 and 33 GB of memory for the two programs.

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